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Module 2 — Excel Fundamentals and Functions

Learning outcomes

Students will be able to:

- identify workbooks, worksheets, rows, columns, cells, ranges, and the formula bar;
- distinguish numbers, text, dates, logical values, blanks, and errors;
- create simple formulas;
- use relative, absolute, and mixed references;
- use common built-in Excel functions; and
- explain zero, errors, and a simple iterative process.

100-minute sequence

Minutes	Activity
0–10	Interface, cell addresses, and data types
10–28	Formulas and relative/absolute references
28–45	Volume-and-cost table demonstration
45–60	SUM, AVERAGE, MIN, MAX, COUNT, and ROUND
60–72	Zero, blank cells, text, and error values
72–84	A simple iterative calculation
84–97	Three-question short exercise
97–100	Review and checklist

Minimum product: one volume-and-cost table whose formulas copy correctly, a function summary, and a five-step iteration table.

1. Main Excel components

- **Workbook:** an Excel file.
- **Worksheet:** a sheet inside a workbook.
- **Cell:** an intersection of a row and column, such as B3.
- **Range:** a group of cells, such as B3:D10.

- **Formula bar:** displays the cell's actual content.
- **Name box:** displays or accepts a cell/range address.

A cell contains a value or formula. Formatting changes its display, not its underlying value. For example, 0.15 displayed as 15% remains numerically 0.15.

2. Data types

Content	Type	Note
12.5	number	can be calculated
Channel A	text	label or code
26/08/2026	date	stored as a date serial number
TRUE	logical	true/false
an unfilled cell	blank	different from zero
#DIV/0!	error	invalid operation, such as division by zero

Do not enter a value as 12 m. Store 12 as a number and put (m) in the column heading.

3. Simple formulas

Every formula begins with =.

=B2*C2

=(B2+C2)/2

=PI()*B2^2/4

Use parentheses to communicate the intended order of operations.

4. Cell references

Assume column B contains volumes and cell F1 contains a unit price.

Formula	Reference type	After copying down one row
=B2*F1	relative	=B3*F2
=B2*\$F\$1	F1 absolute	=B3*\$F\$1
=\$B2*F\$1	mixed	column B and row 1 remain fixed

While editing a cell reference, press **F4** to cycle through F1, \$F\$1, F\$1, and \$F1. F5 opens *Go To*; it does not toggle absolute references.

5. Practice — concrete volume and cost

Create this table:

A	B	C	D	E
Segment	Length (m)	Width (m)	Height (m)	Volume (m ³)
S1	10	0.2	0.3	formula
S2	12	0.2	0.3	formula

Place the concrete unit price in H2. In column F calculate cost:

=E2*\$H\$2

Copy the formula down. Confirm that the volume address changes while H2 remains fixed.

6. Built-in functions

Functions accept inputs, or **arguments**, and return a result. Build a summary for E2:E11:

```
=SUM(E2:E11)
=AVERAGE(E2:E11)
=MIN(E2:E11)
=MAX(E2:E11)
=COUNT(E2:E11)
=ROUND(AVERAGE(E2:E11),3)
```

COUNT counts numeric cells; COUNTA counts nonblank cells. The distinction matters when a dataset contains text or labels.

7. Zero, blanks, and errors

- Zero is a valid number, but may make division undefined.
- A blank cell means data has not been supplied; it should not always be treated as zero.
- Text that looks like a number may disrupt calculations.
- #DIV/0!, #VALUE!, #NAME?, #REF!, and #N/A provide clues about the problem.

Do not hide every error immediately. Identify its cause. Use IFERROR only after deliberately defining the replacement behaviour.

8. Iteration example

Iteration repeats a rule until the result is sufficiently stable. To approximate $\sqrt{2}$, use:

$x_{\text{new}} = (x_{\text{old}} + 2/x_{\text{old}}) / 2$

Start at $x_0 = 1$:

Iteration	old x	new x	absolute change
1	1	$=(B2+2/B2)/2$	$=ABS(C2-B2)$

The new value becomes the old value on the next row. Stop when the change is smaller than a tolerance such as 0.000001. Avoid enabling workbook iteration before understanding circular references and stopping conditions.

Short exercise — 3 questions

1. Predict

Cell C2 contains =A2*\$B\$1. What does it become when copied to C5? What would happen without the dollar signs?

2. Fix

The velocity formula =B2/C2 returns #DIV/0!. Write a check that distinguishes a blank C2, C2=0, and C2>0. Do not simply mask every case with IFERROR.

3. Explain

Starting from $x_0=1$, calculate the first two iterations for the square root of two. Explain why the process needs a tolerance and stopping rule.

1. It becomes `=A5*$B$1`. Without \$, B1 also shifts, becoming B4.
2. Example: `=IF(C2="", "Area is missing", IF(C2=0, "Area cannot be zero", B2/C2))`.
3. $x_1=1.5$; $x_2 \approx 1.4166667$. The tolerance defines when change is small enough; without a stopping condition, iteration may continue indefinitely or create an uncontrolled circular reference.

Checklist

- ☐ I distinguish values, formatting, and formulas.
- ☐ I choose relative or absolute references deliberately.
- ☐ I distinguish zero, blank, text, and errors.
- ☐ I can state an iterative rule and its stopping condition.

Further reading

1. Michael Alexander & Dick Kusleika, *Microsoft Excel 365 Bible*, 2nd ed., Wiley, 2025.
2. Bernard Liengme & Keith Hekman, *Liengme's Guide to Excel 2016 for Scientists and Engineers*, Academic Press, 2019.
3. Ronald W. Larsen, *Engineering with Excel*, 5th ed., Pearson, 2017.
4. Wayne L. Winston, *Microsoft Excel Data Analysis and Business Modeling*, 6th ed., Microsoft Press, 2019.

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